Aaron Sierra

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CAREER PROFILE

Highly motivated software engineer who takes full-stack, AI-powered applications from prototype to production: tool-using LLM agents, model evaluation, and the clean, layered architecture beneath them. Experienced in secure defense environments and in leading small teams. Bilingual (English/Spanish).

AI & LLM: Claude API (tool use, agentic loops, structured outputs, prompt caching), Model Context Protocol (MCP), OpenAI Codex, multi-agent orchestration, model evaluation, MediaPipe, PaddleOCR, TensorFlow/Keras
Languages: Python, Rust, Java, C#/.NET, JavaScript, Dart, SQL, Bash, PowerShell, C/C++
Frameworks & Platforms: FastAPI, Flask, React/Next.js, Spring Boot, .NET MAUI, Docker, Kubernetes, MySQL, SQL Server

EXPERIENCE

L3Harris | Fortune 209 – Waco, TX
Junior Software Engineer **June 2025 – Present**

- Grew daily AI use on a 16-person team from 3 people to all 16 by presenting on prompting and building custom agents; use *OpenAI Codex* and a multi-agent workflow daily to plan, implement, and verify changes.
- Shipped and own a *Python (PyQt)* client for the *JFrog Artifactory REST API* that 30+ people use daily on an air-gapped network, cutting the time of the workflow it replaced by ~70%.
- Refactored a 28-build *Java/Gradle* codebase to run consistently across operating systems, standardizing build scripts.

General Motors | Fortune 25 – Austin, TX
Software Engineer Intern **May 2024 – Aug 2024**

- Led a 5-person team as *Lead Software Engineer* on GM's Operating Systems Engineering team, delivering 3 apps on an internship scoped for 1 and finishing a 25+ story-point plan 3 weeks early; co-presented to CEO Mary Barra.
- Owned the backends of 3 legacy workflow tools, refactoring tangled code into clean, maintainable services with KISS and GRASP principles and porting them to *C#/.NET MAUI* for Windows, macOS, iOS, and Android.
- Delivered tools the 30+ person team and its leadership still use to monitor daily work and adjust workloads: vacation approvals 95% faster, task visibility up 30%, and a coworker directory of 15+ *Azure AD* fields via *Microsoft Graph*, secured with *HashiCorp Vault*.

PROJECTS

AI Golf Swing Coach | github.com/MildTreeRat/golf-coach **Mar 2026 – Present**

- Built an AI coaching app used by 5 golfers, now on iPhone via *Flutter* over the same *Rust* core; a *Claude* agent answers follow-up questions through 12 tools on a custom *Rust MCP server*.
- Engineered it to fail safe: tool loops are bounded, every expected LLM failure (no key, rate limit, refusal) degrades to a note instead of an error, OCR reads arrive flagged, and *structured outputs* make the model refuse rather than guess.
- Built evaluation and cost/latency logging for the app's *Claude* features, grading spec lookups against published specs and coaching via *LLM-as-judge*; across 4 models, found one 60% cheaper at equal accuracy.
- Designed a layered architecture, documented in 35 decision records, that let the core move from Python to 14 *Rust* crates with zero output differences, verified by 3,400+ automated tests.

Study Buddy Finder | Project Manager, 7-person team **Aug 2023 – Dec 2023**

- Led planning and delivery of a full-stack study-partner app (*Next.js/React*, *Spring Boot*, *MySQL*, *Docker Compose*) built through heavy pair programming: 50 REST endpoints, 14 pages, 270+ commits.
- Shipped a ranked matching query that suggests partners and tutors by course, study method, and preferred study time, plus JWT auth with brute-force rate limiting and scheduled session notifications.

Multi-Agent Engineering Workflow | Claude Code, OpenAI Codex **Aug 2026 – Present**

- Created an orchestrator that has run 100+ plan phases unattended, giving each phase a fresh-context subagent gated on automated checks, alongside planning, test-requirements, and design-review agents.

Drug-Molecule GAN | Senior Capstone, 6-person team **Jan 2025 – May 2025**

- Built the *Flask* app and *Docker/Kubernetes* deployment for the team's *TensorFlow* GAN; co-wrote the paper.

EDUCATION

Baylor University – B.S. Computer Science | Concentration in Cybersecurity **May 2025**
Leadership & Certifications: Founding Officer and Mentor, Cyber@Baylor · 3x CCDC Regional Qualifier · CompTIA Security+